

NFC GR Branch Successor Pilot

Generated governance shell

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This generated packet is an L2 branch-pilot shell. It is not a supersession of the canonical GR Branch Book. It tests whether structured claim records and human-authored prose fragments can regenerate a governance-faithful branch packet.

Pilot posture

Computed GR posture: **domain-bounded-cert-close**. The surviving global frontier is **ob:GR.global-subcritical**

Table 1: GR successor-pilot claim catalog

ID	Kind	Declared	Computed	Scope
thm:VI.9.1	theorem	C	C	conditional- certified-regime
thm:GR.CP.1	theorem	C	C	domain-bounded
ob:GR.real	obligation	C	C	domain-bounded
thm:GR.realization	theorem	C	C	domain-bounded
ob:GR.deform	obligation	C	C	domain-bounded
thm:GR.deformation	theorem	C	C	domain-bounded
ob:GR.compat	obligation	C	C	domain-bounded
thm:GR.compat	theorem	C	C	domain-bounded
ob:GR.EFE1	obligation	C	C	domain-bounded
ob:GR.EFE2	obligation	C	C	domain-bounded
ob:GR.EFE3	obligation	C	C	domain-bounded
ob:GR.closure	obligation	C	C	domain-bounded
thm:GR.domain	theorem	C	C	domain-bounded
ob:GR.global-subcriticality	obligation	O	O	global
status:GR	status_note	domain- bounded- cert-close	domain- bounded- cert-close	branch-wide

Generated claim shells

Theorem-like claim: Effective Continuum Legitimacy Theorem (Thm. VI.9.1)

Machine ID. `thm:VI.9.1`

Declared status. C Computed status. C

Scope. regime=certified-interface-regime; domain=spine; certificate_scope=conditional-certified-regime; promotion_ceiling=C; constraints=certified-observable-family,scoped-certificate,continuum-bridge

Effective Continuum Legitimacy Theorem (Thm. VI.9.1) [C]: for certified stabilized observables in a lawful continuum interface regime, differential and integral notation are lawful if and only if they faithfully encode the asymptotic behavior of scale-normalized increments and certified accumulators, respectively, within that declared regime. Proof: the differential direction is thm:VI.differential-legitimacy; the integral direction is thm:VI.integral-legitimacy. Together they give the full biconditional. This is the canonical theorem cited as Thm. VI.9.1 in the GR Branch Book (import list item 5) and referenced in the NS Branch Book (Section 7.7: H^1_x -norm legitimacy). Conditional on: certified observable family, scoped certificate, continuum bridge for the relevant regime.

Dependencies. thm:VI.governing

Theorem-like claim: GR Realization and Coherence Packet (legacy compatibility bundle)

Machine ID. thm:GR.CP.1

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Conditional GR realization/coherence packet placeholder. In a live canon build, this claim would be one basis for reconciling older [O] posture language against later [C] status.

Dependencies. thm:GR.realization, thm:GR.deformation, thm:GR.compat

Proof stub. Proof placeholder.

Obligation: Controlled Causal-Geometric Realization

Machine ID. ob:GR.real

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Controlled Causal-Geometric Realization Migrated GR pilot obligation for the five-stage closure chain. Effective status is checked through explicit discharge edges.

Theorem-like claim: Controlled Causal-Geometric Realization Theorem

Machine ID. thm:GR.realization

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Controlled Causal-Geometric Realization Theorem GR pilot theorem record. The prose stub preserves the human-authored layer while the metadata carries the governance structure.

Dependencies. thm:VI.9.1

Discharge edges. ob:GR.real

Proof stub.

Proof stub for Controlled Causal-Geometric Realization Theorem The v11 pilot records only the governance scaffold; full mathematical proof text remains in the canonical branch source.

Obligation: Causal-Deformation Response Law

Machine ID. ob:GR.deform

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Causal-Deformation Response Law Migrated GR pilot obligation for the five-stage closure chain. Effective status is checked through explicit discharge edges.

Dependencies. ob:GR.real

Theorem-like claim: Causal-Deformation Law Theorem

Machine ID. thm:GR.deformation

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Causal-Deformation Law Theorem GR pilot theorem record. The prose stub preserves the human-authored layer while the metadata carries the governance structure.

Dependencies. thm:GR.realization

Discharge edges. ob:GR.deform

Proof stub.

Proof stub for Causal-Deformation Law Theorem The v11 pilot records only the governance scaffold; full mathematical proof text remains in the canonical branch source.

Obligation: Metric/Connection and Einstein-Type Structural Compatibility

Machine ID. ob:GR.compat

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Metric/Connection and Einstein-Type Structural Compatibility Migrated GR pilot obligation for the five-stage closure chain. Effective status is checked through explicit discharge edges.

Dependencies. ob:GR.deform

Theorem-like claim: Metric/Connection Compatible Structural Compatibility Packet

Machine ID. thm:GR.compat

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Metric/Connection Compatible Structural Compatibility Packet GR pilot theorem record. The prose stub preserves the human-authored layer while the metadata carries the governance structure.

Dependencies. thm:GR.deformation

Discharge edges. ob:GR.compat, ob:GR.EFE1, ob:GR.EFE2, ob:GR.EFE3

Proof stub.

Proof stub for Metric/Connection Compatible Structural Compatibility Packet The v11 pilot records only the governance scaffold; full mathematical proof text remains in the canonical branch source.

Obligation: EFE Curvature Encoding Obligation

Machine ID. ob:GR.EFE1

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Open obligation placeholder for Einstein-form structural compatibility.

Dependencies. thm:GR.compat

Obligation: EFE Coupling Constants Obligation

Machine ID. ob:GR.EFE2

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

EFE Coupling Constants Obligation Migrated GR pilot obligation. This obligation is treated as conditionally discharged at domain-bounded scope by the GR compatibility packet.

Dependencies. thm:GR.compat

Obligation: EFE Diffeomorphism Invariance Obligation

Machine ID. ob:GR.EFE3

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

EFE Diffeomorphism Invariance Obligation Migrated GR pilot obligation. This obligation is treated as conditionally discharged at domain-bounded scope by the GR compatibility packet.

Dependencies. thm:GR.compat

Obligation: Domain-Bounded Gravity Closure Package

Machine ID. ob:GR.closure

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Domain-Bounded Gravity Closure Package Migrated GR pilot obligation for the five-stage closure chain. Effective status is checked through explicit discharge edges.

Dependencies. ob:GR.compat, ob:GR.EFE1, ob:GR.EFE2, ob:GR.EFE3

Theorem-like claim: Domain-Bounded Gravity Closure Theorem

Machine ID. thm:GR.domain

Declared status. C **Computed status.** C

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=domain-bounded; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D

Domain-Bounded Gravity Closure Theorem GR pilot theorem record. The prose stub preserves the human-authored layer while the metadata carries the governance structure.

Dependencies. thm:GR.compat

Discharge edges. ob:GR.closure

Proof stub.

Proof stub for Domain-Bounded Gravity Closure Theorem The v11 pilot records only the governance scaffold; full mathematical proof text remains in the canonical branch source.

Obligation: Global Curvature-Subcriticality / BF-9 Extension Frontier

Machine ID. ob:GR.global-subcriticality

Declared status. O **Computed status.** O

Scope. regime=declared-GR-target; domain=GR-branch; certificate_scope=global; continuum_license=book-VI-certified; promotion_ceiling=C; constraints=realized-domain-D,BF-9-extension-required

Global curvature-subcriticality remains the GR frontier beyond realized-domain closure.

Dependencies. thm:GR.domain

Status note: GR Branch Posture Record

Machine ID. status:GR

Declared status. domain-bounded-cert-close **Computed status.** domain-bounded-cert-close

Scope. regime=gr-branch-governance; domain=GR-branch; certificate_scope=branch-wide; continuum_license=book-VI-certified; promotion_ceiling=domain-bounded-cert-close; constraints=realized-domain-D

Older local posture language still says CERT-PROJ although the branch has domain-bounded closure in the reconciled state sheet.

Dependencies. ob:GR.global-subcriticality, thm:GR.CP.1, thm:GR.domain, thm:VII.6.4, thm:V.UBLT